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STANDARD OPERATING PROCEDURE

Design, Engineering, Manufacturing and Supply of Bridge Construction Equipment, Formwork and False work Fabrication

Applicable Standards:

IS 7307:2016, IS 9595:1980, ISO 9001:2015

Scope:

the complete workflow for design, engineering, manufacturing, inspection, surface treatment, assembly, supply, and after-sales support for bridge construction equipment, formwork, false work, Movable scaffolding, and fabricated structural systems.

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1. PURPOSE

The purpose of this SOP is to establish a systematic procedure for the design, engineering, manufacturing, quality control, inspection, packing, transportation, and supply of bridge construction equipment and fabricated systems in compliance with applicable Indian Standards and quality management requirements.

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2. OBJECTIVE

- Ensure product quality, safety, and reliability.
- Maintain compliance with IS 7307:2016, IS 9595:1980, and ISO 9001:2015.
- Standardize manufacturing and engineering processes.
- Ensure timely delivery and customer satisfaction.
- Maintain traceability and documentation throughout the project lifecycle.

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3. APPLICABLE STANDARDS

Standard	Description
IS 7307:2016	Code of Practice for Construction Equipment
IS 9595:1980	Specification for Steel Scaffolding and Tubular Structures
ISO 9001:2015	Quality Management System
Approved WPS	Welding Procedure Specifications
Approved Drawings	Client Approved Drawings and Documents

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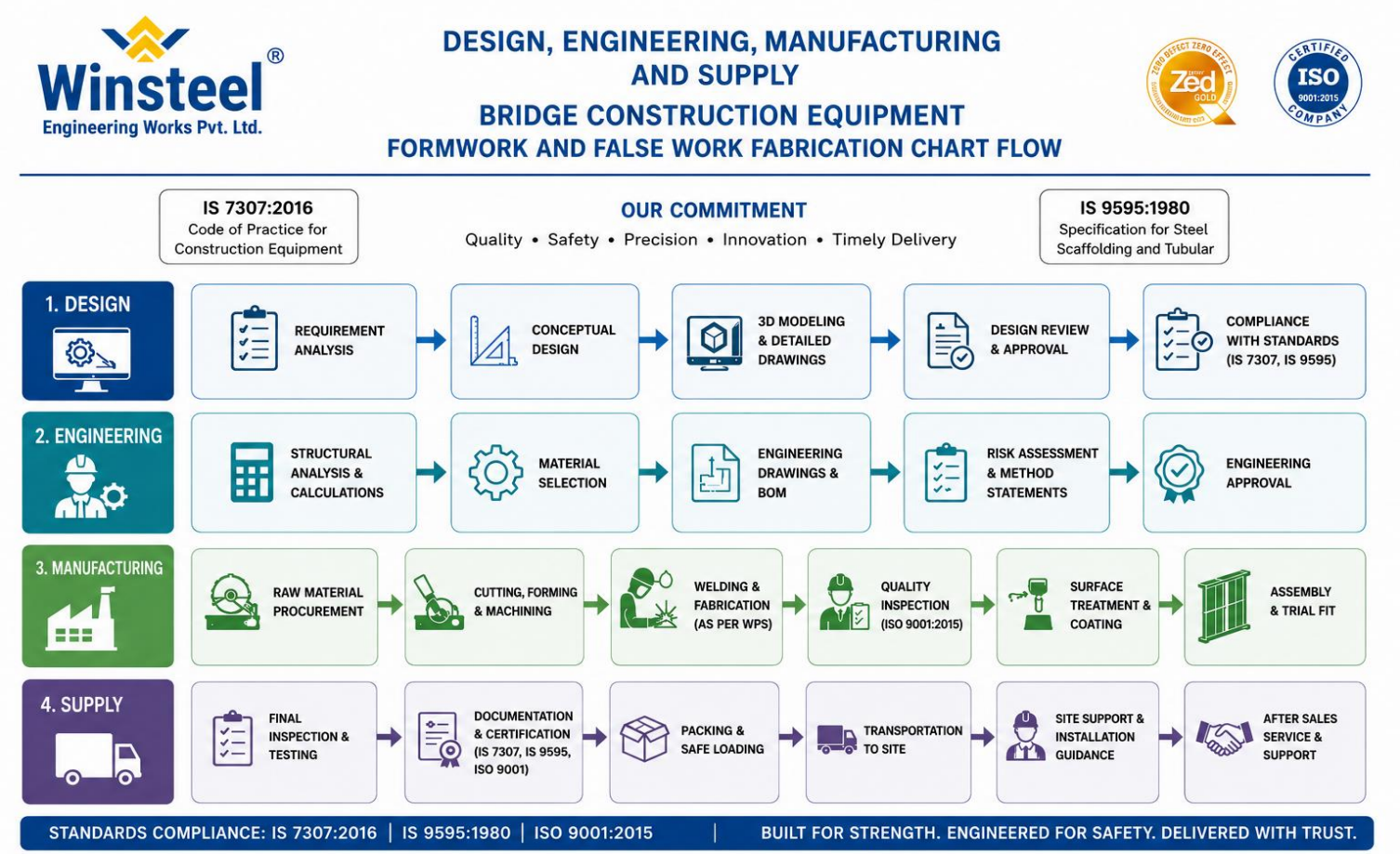
4. OUR COMMITMENT

- Quality
- Safety
- Precision
- Innovation
- Timely Delivery

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5. PROCESS FLOW OVERVIEW

1-Design → 2-Engineering → 3-Manufacturing → 4-Supply



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6. DESIGN PROCEDURE

6.1 Requirement Analysis

Activities

- Review client specifications and tender drawing.
- Understand project scope, loading conditions, geometry, and site constraints.
- Identify design codes and standards.
- Conduct feasibility review.

Responsibility

- Design Department
- Project Coordinator

Documents

- Client specifications
- Design basis report
- Technical inputs

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6.2 Conceptual Design

Activities

- Develop conceptual layouts and preliminary schemes.
- Select suitable construction methodology.
- Evaluate stability and constructability.

Responsibility

- Design Engineers

Deliverables

- Concept drawings
- Preliminary calculations
- Concept approval documents

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6.3 3D Modeling & Detailed Drawings

Activities

- Prepare 3D models and fabrication drawings.
- Generate GA drawings, shop drawings, and assembly details.
- Verify dimensions and interfaces.

Responsibility

- CAD Team
- Design Department

Software

- AutoCAD
- Tekla
- SolidWorks or equivalent

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6.4 Design Review & Approval

Activities

- Conduct internal design review.
- Verify calculations and safety factors.
- Obtain client approval before manufacturing.

Responsibility

- Design Head
- QA/QC Department
- Client Representative

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6.5 Compliance Verification

Activities

- Ensure compliance with IS 7307 and IS 9595.
- Confirm structural adequacy and safety.
- Review all revisions and approvals.

Records

- Approved drawings
- Design calculation sheets
- Compliance reports

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7. ENGINEERING PROCEDURE

7.1 Structural Analysis & Calculations

Activities

- Perform structural analysis for dead load, live load, wind load, and erection load.
- Validate member capacities and stability.
- Prepare calculation reports.

Responsibility

- Structural Engineers

Deliverables

- Structural calculation report
- Load analysis sheets

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7.2 Material Selection

Activities

- Select material grades based on design requirements.
- Verify material standards and availability.
- Ensure traceability of raw materials.

Responsibility

- Engineering Department
- Procurement Team

Typical Materials

- Structural Steel
- Plates
- Pipes
- Fasteners

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7.3 Engineering Drawings & BOM

Activities

- Prepare detailed fabrication drawings.
- Generate Bill of Materials (BOM).
- Issue controlled drawings for production.

Responsibility

- Engineering Department

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7.4 Risk Assessment & Method Statement

Activities

- Identify project risks and mitigation measures.
- Prepare erection and fabrication method statements.
- Conduct safety review.

Responsibility

- Safety Department
- Project Team

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7.5 Engineering Approval

Activities

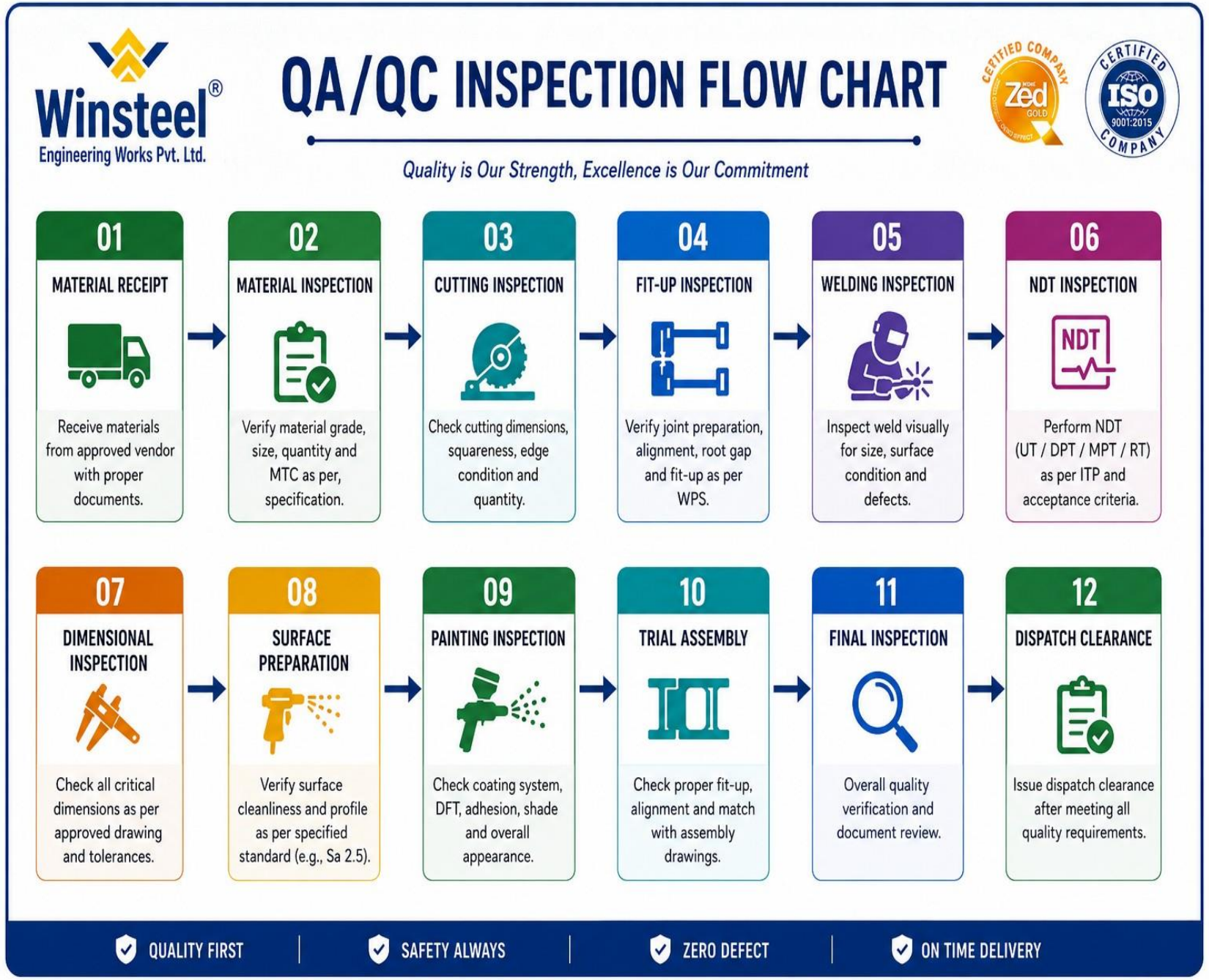
- Final engineering review.
- Client approval and release for manufacturing.

Records

- Approved engineering documents
- Method statements
- Inspection plans

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8. MANUFACTURING PROCEDURE

8.1 Raw Material Procurement

Activities

- Procure approved raw materials from qualified vendors.
- Verify Mill Test Certificates (MTC).
- Conduct incoming inspection.

Responsibility

- Procurement Department
- QA/QC Department

Inspection

- Visual inspection
- Dimensional verification
- Material traceability check

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8.2 Cutting, Forming & Machining

Activities

- Perform cutting using approved equipment.
- Carry out bending, drilling, and machining.
- Ensure dimensional accuracy.

Equipment

- CNC cutting machines
- Drilling machines
- Hydraulic presses

Quality Control

- Dimensional inspection
- Fit-up verification

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8.3 Welding & Fabrication

Activities

- Welding shall be performed as per approved WPS.
- Qualified welders shall carry out fabrication.
- Maintain welding consumable control.

Standards

- Approved WPS/PQR
- Relevant welding codes

Inspection

- Visual inspection
- Weld quality verification
- NDT where applicable

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8.4 Quality Inspection

Activities

- Perform stage-wise inspection.
- Verify dimensions, weld quality, and fitment.
- Ensure ISO 9001:2015 compliance.

Responsibility

- QA/QC Department

Inspection Records

- Inspection reports
- Calibration records
- Checklists

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8.5 Surface Treatment & Coating

Activities

- Surface preparation by approved mechanical cleaning methods.
- Primer application immediately after cleaning.
- Final painting as per approved DFT requirements.

Requirements

- DFT as per approved specification.
- Approved color shade: Win-Blue and Yellow.

Inspection

- Surface cleanliness check
- DFT measurement
- Visual coating inspection

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8.6 Assembly & Trial Fit

Activities

- Conduct trial assembly of components.
- Verify alignment, dimensions, and functionality.
- Resolve fit-up issues before dispatch.

Responsibility

- Production Department
- QA/QC Team

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9. SUPPLY PROCEDURE

9.1 Final Inspection & Testing

Activities

- Conduct final dimensional inspection.
- Verify quality records and approvals.
- Perform load testing where applicable.

Responsibility

- QA/QC Department
- Client Inspector (if required)

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9.2 Documentation & Certification

Documents

- Inspection reports
- Material Test Certificates
- Welding records
- Paint inspection reports
- Compliance certificates
- ISO certificates

Standards

- IS 7307
- IS 9595
- ISO 9001:2015

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9.3 Packing & Safe Loading

Activities

- Use suitable packing materials.
- Protect components against damage during transportation.
- Ensure safe loading practices.

Responsibility

- Dispatch Department

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9.4 Transportation to Site

Activities

- Arrange transportation as per dispatch plan.
- Ensure safe handling during transit.
- Track delivery schedule.

Responsibility

- Logistics Department

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9.5 Site Support & Installation Guidance

Activities

- Provide erection guidance.
- Assist with installation procedures.
- Address technical queries at site.

Responsibility

- Site Support Team
- Project Engineers

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9.6 After Sales Service & Support

Activities

- Provide post-delivery technical support.
- Attend client concerns and corrective actions.
- Maintain customer feedback records.

Responsibility

- Service Department

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10. QUALITY CONTROL REQUIREMENTS

- All inspections shall be documented.
- Measuring instruments shall be calibrated.
- Full traceability shall be maintained.
- Welding consumables shall be controlled.
- Non-conformance shall be recorded and corrected.

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11. SAFETY REQUIREMENTS

- PPE shall be mandatory in all production areas.
- Hot work permits shall be followed.
- Lifting tools and tackles shall be certified.
- Safe material handling practices shall be followed.
- Emergency procedures shall be displayed.

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12. RECORDS TO BE MAINTAINED

- Approved drawings
- Structural calculations
- BOM records
- Material Test Certificates
- Welding records
- Inspection reports
- Calibration certificates
- Paint inspection reports
- Dispatch records
- Client approvals

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13. RESPONSIBILITY MATRIX

Department	Responsibility
Design	Concept and detailed design
Engineering	Structural calculations and engineering approvals
Procurement	Material sourcing and vendor coordination
Production	Fabrication and assembly
QA/QC	Inspection and quality assurance
Safety	Safety implementation and monitoring
Dispatch	Packing and transportation
Site Team	Installation support

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14. REVISION CONTROL

Revision	Description	Date
00	Initial Release	As Applicable

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15. CONCLUSION

This SOP establishes a comprehensive and standardized approach for the successful execution of bridge construction equipment, formwork, false work, scaffolding, and fabricated structural projects while ensuring compliance with applicable standards, quality requirements, and client expectations.

Built for Strength. Engineered for Safety. Delivered with Trust.

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THANK YOU

